

Adversarial Resilience



[Your Name]
[Date]

How to Use this Template

- We have provided these slides as a guide to ensure you submit all the required components to complete your project successfully.
- When presenting your project, remember that these slides are merely a guide. We strongly encourage you to embrace your creative freedom and make changes that reflect your unique vision as long as the required information is present.
- You can add slides to the template when your answers or screenshots do not fit on the previously provided pages.
- Delete this and the next slide before submitting your project.
- **Remember to add your name and the date to the cover page.**

Project Scenario

Your employer, NuttyUtility, is a technology company with cutting-edge software that is known to be targeted by foreign adversaries. It has recently acquired a new company, StaticSpeed. Your job is to assess the security posture of the machines in the newly acquired company.

Upon our initial inspection during the acquisition, it became apparent that the systems at StaticSpeed had not been adequately maintained. There is a high probability that the servers and desktops are running outdated applications and misconfigurations, which could potentially lead to breaches by either internal or external malicious actors. This situation is unacceptable, especially considering our plans to integrate StaticSpeed's systems with our extended network.

Your task is to perform a security assessment of the provided Windows and Linux virtual machines, which the company uses as golden images. The result of this assessment will be a report that you must present to your stakeholders. They will relay your findings to the infrastructure team. Together, they will decide if this new company is ready to be integrated or if appropriate controls and mitigations need to be put in place before this happens.



Section One: Windows Assessment

CIS Benchmark Compliance Evaluation on Windows

In this section, you'll dive into the heart of Windows security by evaluating specific CIS benchmarks on a Windows virtual machine (VM). This hands-on exercise mirrors real-world scenarios, where you'll need to assess and enhance the security posture of systems critical to NuttyUtility's operations. Your expertise will help ensure these systems are fortified against adversarial threats, safeguarding our cutting-edge software from potential breaches.

Your Task:

You need to evaluate the following CIS benchmarks on the provided Windows VM. For each benchmark:

- Decide if the machine is in compliance with the benchmark.
- Provide screenshot evidence of your compliance decision.
- Provide remediation steps for any benchmarks that the machine does not meet.

Pay attention: When a control is not set, it also means the control is not met. Not every control has a default "disabled" status.



Compliance Evaluation on Windows

Benchmark:	1.1.5 (L1) Ensure 'Password must meet complexity requirements' is set to 'Enabled'
Compliance:	[No]
Remediation strategy:	Enable password complexity requirements through Local Security Policy or Group Policy

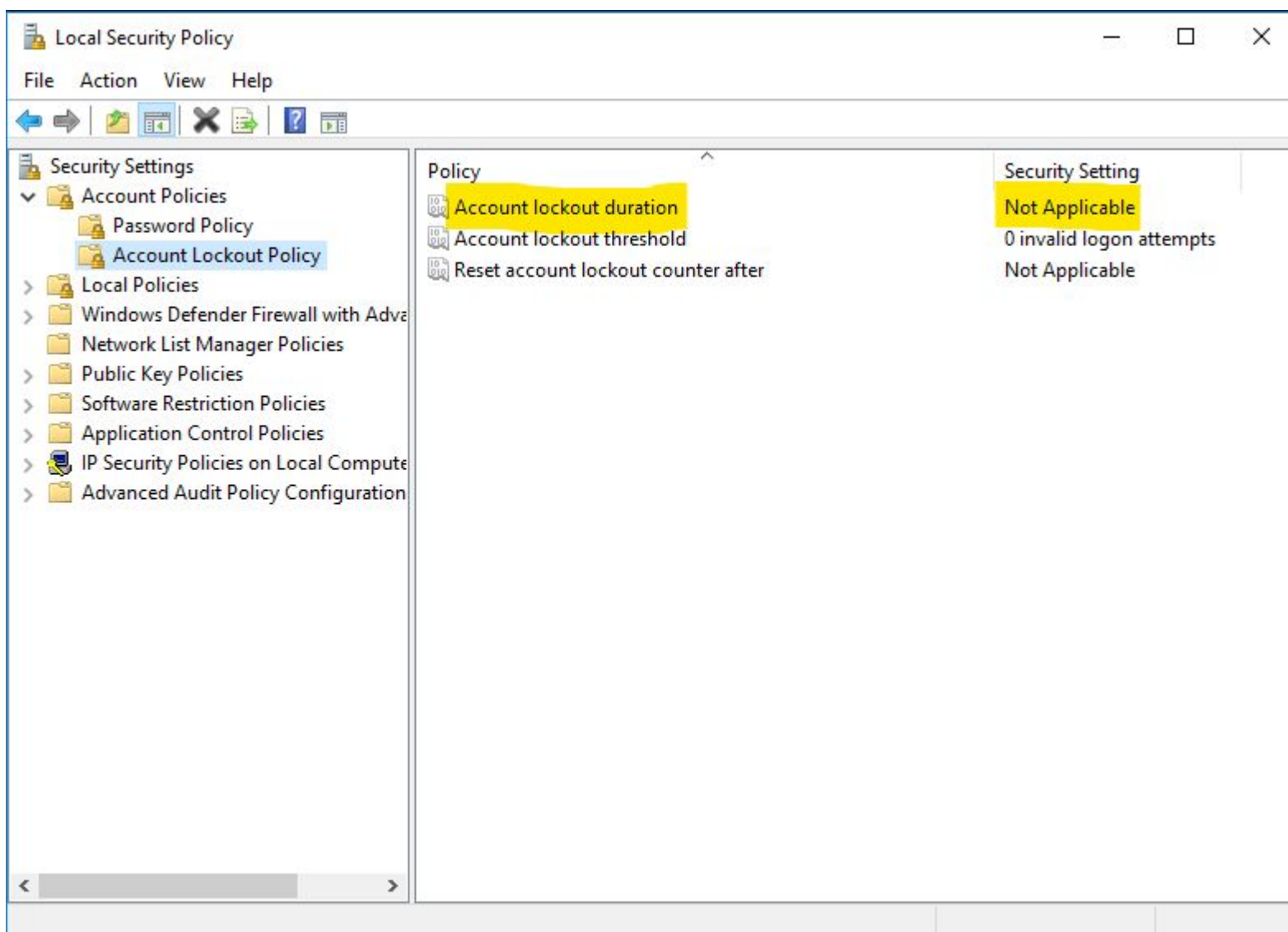
The screenshot shows the 'Local Security Policy' window. In the left-hand tree, 'Security Settings' is expanded, and 'Account Policies' > 'Password Policy' is selected. The main pane displays a list of password-related policies. The policy 'Password must meet complexity requirements' is highlighted with a yellow background. To its right, the 'Security Setting' column shows its current state: 'Disabled'.

Policy	Security Setting
Enforce password history	0 passwords remembered
Maximum password age	42 days
Minimum password age	0 days
Minimum password length	0 characters
Password must meet complexity requirements	Disabled
Store passwords using reversible encryption	Disabled



Compliance Evaluation on Windows

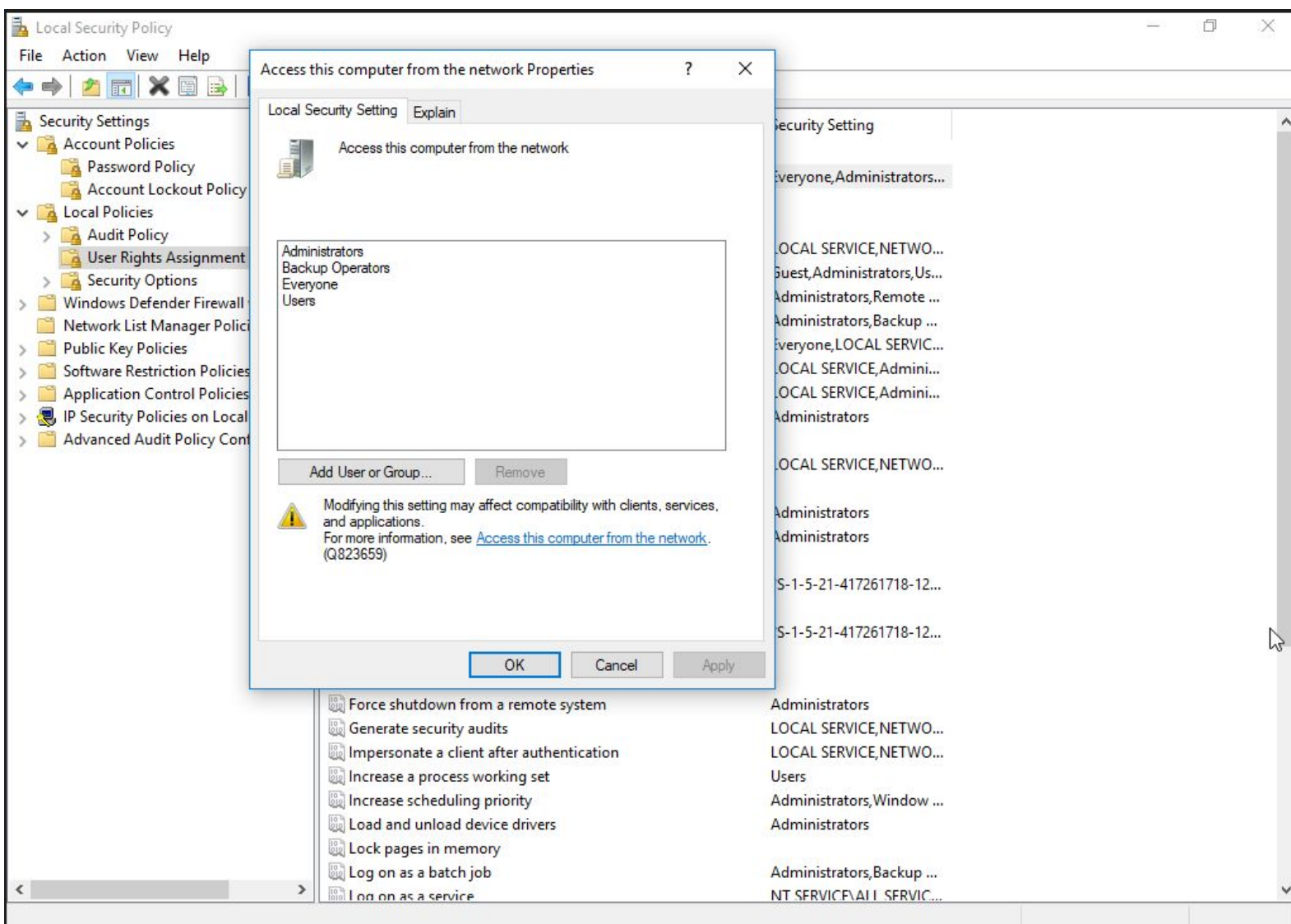
Benchmark:	1.2.1 Ensure 'Account lockout duration' is set to '15 or more minute(s)'
Compliance:	[No]
Remediation strategy:	Configure <i>Account lockout duration</i> to <i>15 minutes</i> or more through Local Security Policy, Group Policy, or command line.





Compliance Evaluation on Windows

Benchmark:	2.2.2 Ensure 'Access this computer from the network' is set to 'Administrators, Remote Desktop Users'
Compliance:	[No]
Remediation strategy:	Configure the policy so that only Administrators and Remote Desktop Users have the Access this computer from the network right.





Compliance Evaluation on Windows

Benchmark:	2.3.1.1 Ensure 'Accounts: Administrator account status' is set to 'Disabled'
Compliance:	[No]
Remediation strategy:	Disable the built-in local Administrator account through Local Security Policy, PowerShell, or command line.

Local Security Policy

File Action View Help

Security Settings

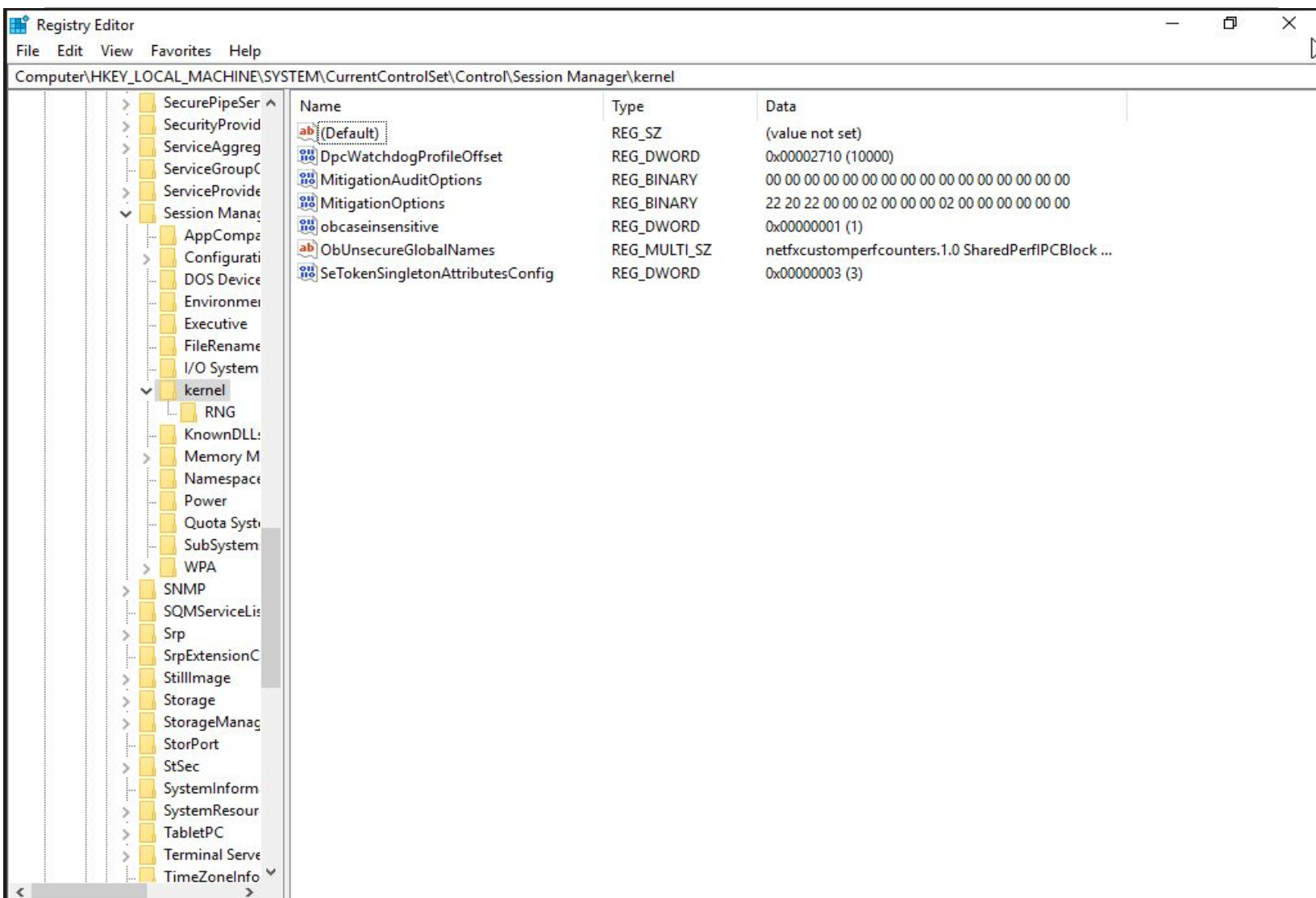
- Account Policies
 - Password Policy
 - Account Lockout Policy
- Local Policies
 - Audit Policy
 - User Rights Assignment
 - Security Options
- Windows Defender Firewall with Advanced Security
- Network List Manager Policies
- Public Key Policies
- Software Restriction Policies
- Application Control Policies
- IP Security Policies on Local Computer
- Advanced Audit Policy Configuration

Policy	Security Setting
Accounts: Administrator account status	Enabled
Accounts: Block Microsoft accounts	Not Defined
Accounts: Guest account status	Enabled
Accounts: Limit local account use of blank passwords to computer administrators	Enabled
Accounts: Rename administrator account	Administrator
Accounts: Rename guest account	Guest
Audit: Audit the access of global system objects	Disabled
Audit: Audit the use of Backup and Restore privilege	Disabled
Audit: Force audit policy subcategory settings (Windows Vista and Windows 7 only)	Not Defined
Audit: Shut down system immediately if unable to log security events	Disabled
DCOM: Machine Access Restrictions in Security Descriptor Database	Not Defined
DCOM: Machine Launch Restrictions in Security Descriptor Database	Not Defined
Devices: Allow undock without having to log on	Enabled
Devices: Allowed to format and eject removable media	Not Defined
Devices: Prevent users from installing printer drivers	Disabled
Devices: Restrict CD-ROM access to locally logged-on user only	Not Defined
Devices: Restrict floppy access to locally logged-on user only	Not Defined
Domain controller: Allow server operators to schedule tasks	Not Defined
Domain controller: LDAP server signing requirements	Not Defined
Domain controller: Refuse machine account password changes	Not Defined
Domain member: Digitally encrypt or sign secure channel data (when possible)	Enabled
Domain member: Digitally encrypt secure channel data (when possible)	Enabled
Domain member: Digitally sign secure channel data (when possible)	Enabled
Domain member: Disable machine account password changes	Disabled
Domain member: Maximum machine account password age	30 days
Domain member: Require strong (Windows 2000 or later) session security	Enabled
Interactive logon: Display user information when the session is reactivated	Not Defined
Interactive logon: Do not require CTRL+ALT+DEL	Not Defined
Interactive logon: Don't display last signed-in	Disabled
Interactive logon: Don't display username at sign-in	Not Defined
Interactive logon: Machine account lockout threshold	Not Defined
Interactive logon: Machine inactivity limit	Not Defined



Compliance Evaluation on Windows

Benchmark:	18.3.4: Ensure 'Enable Structured Exception Handling Overwrite Protection (SEHOP)' is set to 'Enabled'
Compliance:	[No]
Remediation strategy:	Enable SEHOP through Registry by creating DisableExceptionChainValidation key under HKLM\SYSTEM\CurrentControlSet\Control\Session Manager\kernel\DisableExceptionChainValidation and setting the relevant registry value to 0





Compliance Evaluation on Windows

Benchmark:	18.9.102.2: Ensure 'configure automatic updates' is set to 'Enabled.'
Compliance:	[No]
Remediation strategy:	Enable the Configure Automatic Updates Group Policy setting and select an appropriate automatic update behavior

Local Group Policy Editor

File Action View Help

Speech Store Sync your settings Tablet PC Task Scheduler Windows Calendar Windows Color Sys Windows Custome Windows Defender Windows Defender Windows Defender Windows Defender Windows Defender Windows Error Rep Windows Game Re Windows Hello for Windows Ink Work Windows Installer Windows Logon O Windows Media Di Windows Media Pl Windows Messeng Windows Mobility Windows PowerSh Windows Reliability Windows Remote S Windows Update Windows Upda Work Folders All Settings User Configuration Software Settings Windows Settings

Windows Update

Configure Automatic Updates

Edit [policy setting](#)

Requirements:
Windows XP Professional Service Pack 1 or At least Windows 2000 Service Pack 3

Description:
Specifies whether this computer will receive security updates and other important downloads through the Windows automatic updating service.

Note: This policy does not apply to Windows RT.

This setting lets you specify whether automatic updates are enabled on this computer. If the service is enabled, you must select one of the four options in the Group Policy Setting:

2 = Notify before downloading and installing any updates.

When Windows finds updates that apply to this computer, users will be notified that updates are ready to be downloaded. After going to Windows Update, users can download and install any available updates.

Setting	State	Comment
Turn off auto-restart for updates during active hours	Enabled	No
Specify active hours range for auto-restarts	Not configured	No
Allow updates to be downloaded automatically over metered connections	Not configured	No
Always automatically restart at the scheduled time	Disabled	No
Specify deadline before auto-restart for update installation	Not configured	No
Configure auto-restart reminder notifications for updates	Not configured	No
Turn off auto-restart notifications for update installations	Not configured	No
Configure auto-restart required notification for updates	Not configured	No
Configure Automatic Updates	Disabled	No
Specify intranet Microsoft update service location	Not configured	No
Automatic Updates detection frequency	Disabled	No
Do not allow update deferral policies to cause scans against ...	Not configured	No
Remove access to use all Windows Update features	Not configured	No
Do not connect to any Windows Update Internet locations	Enabled	No
Allow non-administrators to receive update notifications	Not configured	No
Specify Engaged restart transition and notification schedule ...	Not configured	No
Do not include drivers with Windows Updates	Not configured	No
Turn on Software Notifications	Disabled	No
Allow Automatic Updates immediate installation	Disabled	No
Turn on recommended updates via Automatic Updates	Disabled	No
No auto-restart with logged on users for scheduled automat...	Not configured	No
Re-prompt for restart with scheduled installations	Not configured	No
Delay Restart for scheduled installations	Not configured	No
Reschedule Automatic Updates scheduled installations	Not configured	No
Configure auto-restart warning notifications schedule for u...	Not configured	No
Update Power Policy for Cart Restarts	Not configured	No
Enable client-side targeting	Not configured	No
Allow signed updates from an intranet Microsoft update ser...	Not configured	No

Extended Standard

31 setting(s)



Section Two: Ubuntu Assessment

CIS Benchmark Compliance Evaluation on Ubuntu 18.04

Here, you'll roll up your sleeves and tackle the security evaluation of an Ubuntu Linux 18.04 VM. This task is designed to reflect real-world challenges, requiring you to scrutinize and uplift the security measures of our systems. By ensuring compliance with CIS benchmarks, you'll play a pivotal role in fortifying StaticSpeed's systems against vulnerabilities, making sure they're robust enough to integrate with NuttyUtility's secure network.

Your Task:

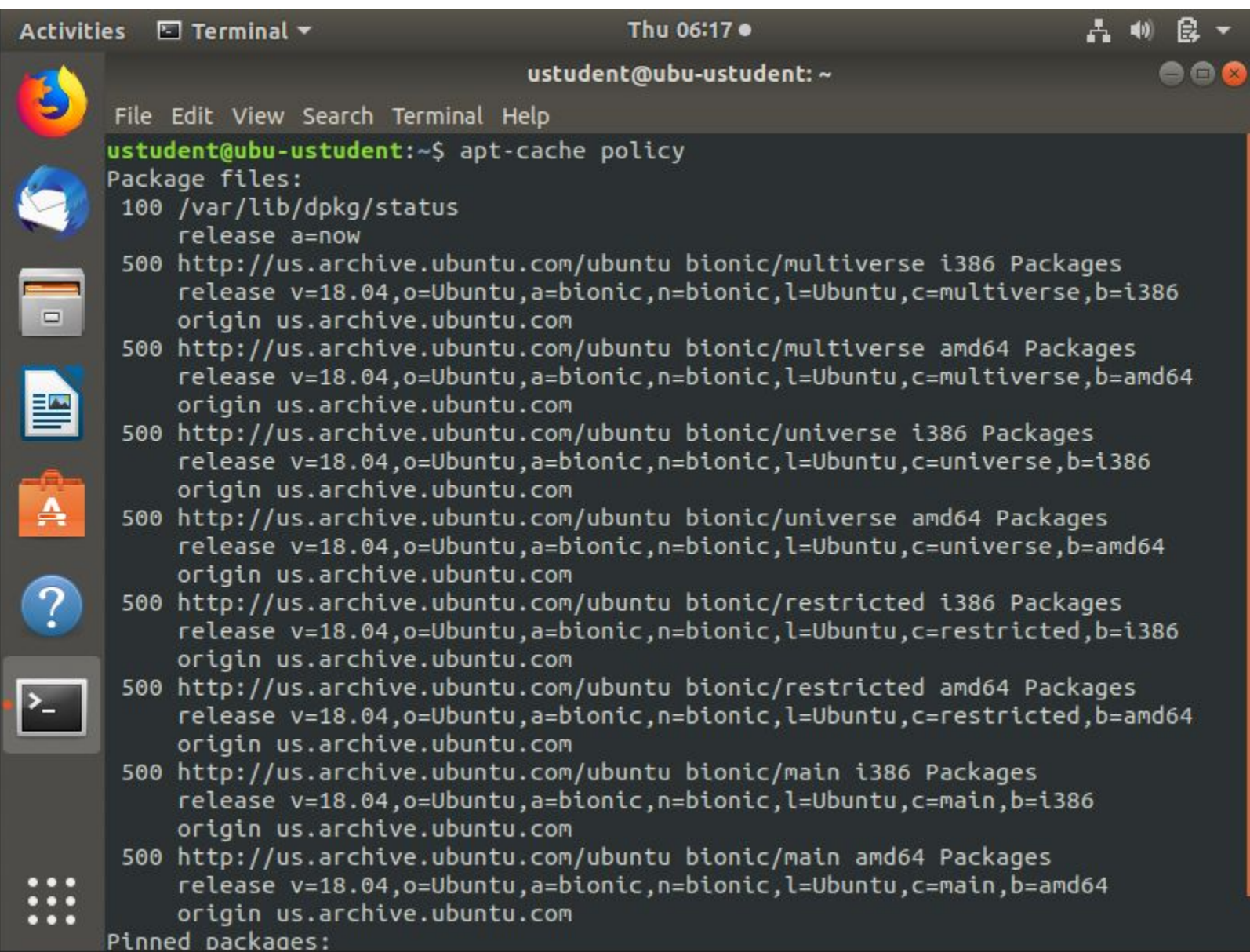
You need to evaluate the following CIS benchmarks on the provided Ubuntu Linux 18.04 VM. For each benchmark:

- Decide if the machine is in compliance with the benchmark.
- Provide screenshot evidence of your compliance decision.
- Provide remediation steps for any benchmarks that the machine does not meet.



Compliance Evaluation on Ubuntu

Benchmark:	1.2.1: <i>Ensure package manager repositories are configured</i>
Compliance:	[Yes]
Remediation strategy:	



```
ustudent@ubu-ustudent: ~$ apt-cache policy
Package files:
100 /var/lib/dpkg/status
   release a=now
500 http://us.archive.ubuntu.com/ubuntu bionic/multiverse i386 Packages
   release v=18.04,o=Ubuntu,a=bionic,n=bionic,l=Ubuntu,c=multiverse,b=i386
   origin us.archive.ubuntu.com
500 http://us.archive.ubuntu.com/ubuntu bionic/multiverse amd64 Packages
   release v=18.04,o=Ubuntu,a=bionic,n=bionic,l=Ubuntu,c=multiverse,b=amd64
   origin us.archive.ubuntu.com
500 http://us.archive.ubuntu.com/ubuntu bionic/universe i386 Packages
   release v=18.04,o=Ubuntu,a=bionic,n=bionic,l=Ubuntu,c=universe,b=i386
   origin us.archive.ubuntu.com
500 http://us.archive.ubuntu.com/ubuntu bionic/universe amd64 Packages
   release v=18.04,o=Ubuntu,a=bionic,n=bionic,l=Ubuntu,c=universe,b=amd64
   origin us.archive.ubuntu.com
500 http://us.archive.ubuntu.com/ubuntu bionic/restricted i386 Packages
   release v=18.04,o=Ubuntu,a=bionic,n=bionic,l=Ubuntu,c=restricted,b=i386
   origin us.archive.ubuntu.com
500 http://us.archive.ubuntu.com/ubuntu bionic/restricted amd64 Packages
   release v=18.04,o=Ubuntu,a=bionic,n=bionic,l=Ubuntu,c=restricted,b=amd64
   origin us.archive.ubuntu.com
500 http://us.archive.ubuntu.com/ubuntu bionic/main i386 Packages
   release v=18.04,o=Ubuntu,a=bionic,n=bionic,l=Ubuntu,c=main,b=i386
   origin us.archive.ubuntu.com
500 http://us.archive.ubuntu.com/ubuntu bionic/main amd64 Packages
   release v=18.04,o=Ubuntu,a=bionic,n=bionic,l=Ubuntu,c=main,b=amd64
   origin us.archive.ubuntu.com
Pinned packages:
```



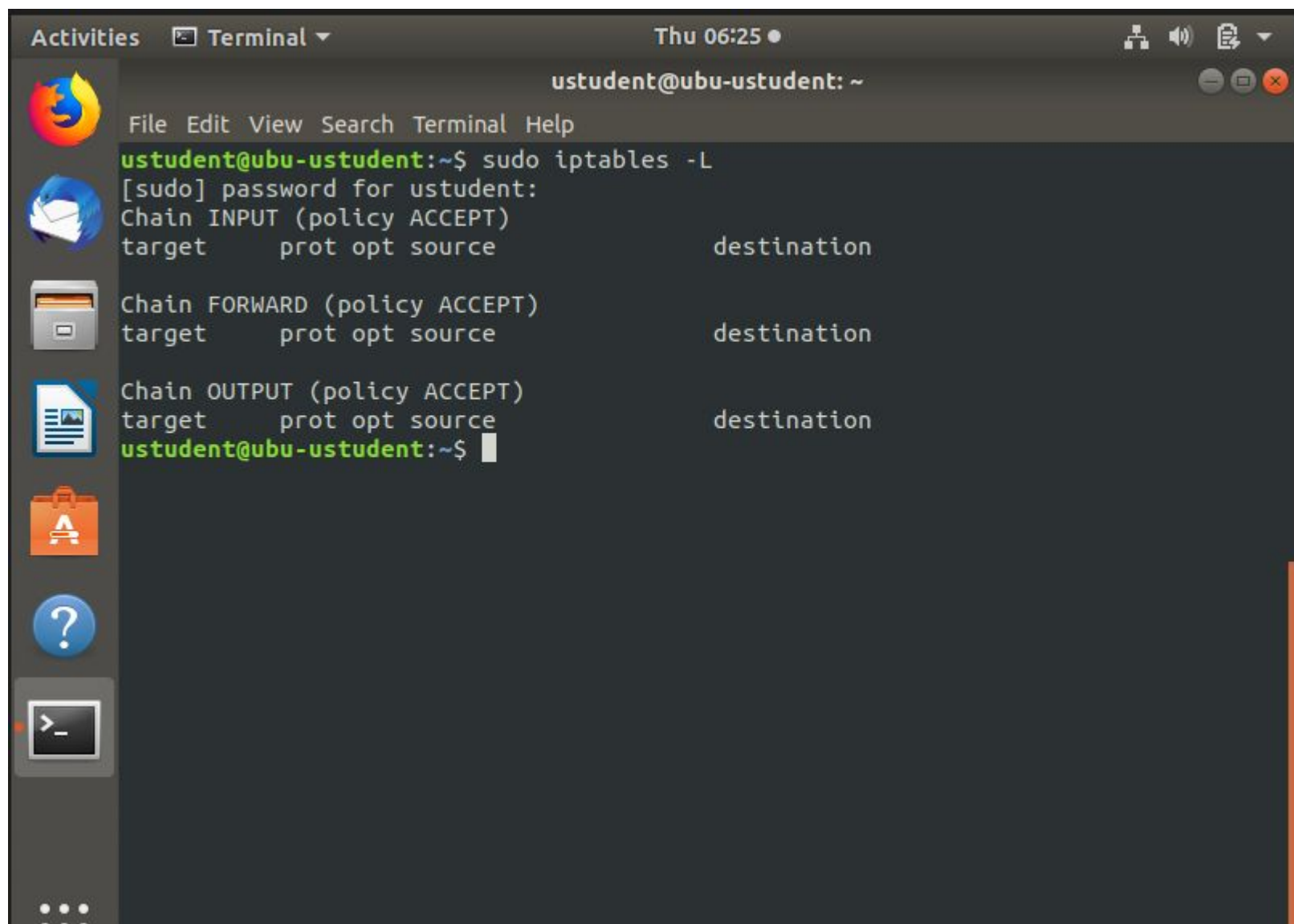
Compliance Evaluation on Ubuntu

Benchmark:	1.6.2 Ensure address space layout randomization (ASLR) is enabled
Compliance:	[Yes]
Remediation strategy:	

```
Activities  Terminal ▾  Thu 06:20 ●  ustudent@ubu-ustudent: ~  
File Edit View Search Terminal Help  
ustudent@ubu-ustudent:~$ sysctl kernel.randomize_va_space  
kernel.randomize_va_space = 2  
ustudent@ubu-ustudent:~$
```


Compliance Evaluation on Ubuntu

Benchmark:	<i>3.5.3.1 Ensure iptables are flushed</i>
Compliance:	<i>[Yes]</i>
Remediation strategy:	



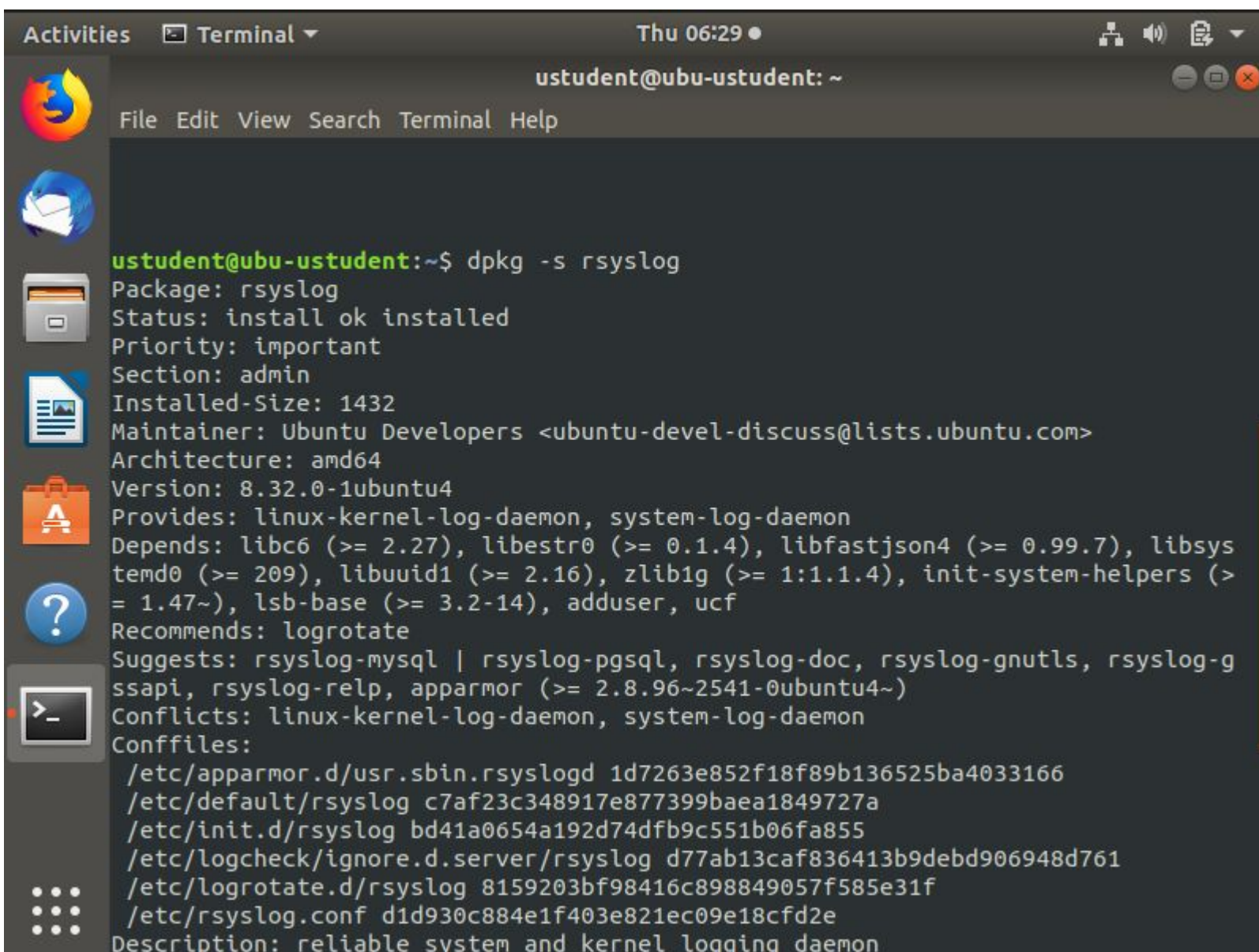
A screenshot of a Linux terminal window titled "Terminal" with a dark theme. The window shows the command `sudo iptables -L` being executed. The output lists three chains: INPUT, FORWARD, and OUTPUT, all with a policy of ACCEPT. Each chain shows a table of rules with columns for target, protocol, options, source, and destination. The terminal window has a sidebar on the left with icons for various applications, and the top bar shows the time as "Thu 06:25".

```
ustudent@ubu-ustudent: ~  
File Edit View Search Terminal Help  
ustudent@ubu-ustudent:~$ sudo iptables -L  
[sudo] password for ustudent:  
Chain INPUT (policy ACCEPT)  
target      prot opt source                destination  
  
Chain FORWARD (policy ACCEPT)  
target      prot opt source                destination  
  
Chain OUTPUT (policy ACCEPT)  
target      prot opt source                destination  
ustudent@ubu-ustudent:~$
```



Compliance Evaluation on Ubuntu

Benchmark:	4.2.1.1 Ensure rsyslog is installed
Compliance:	[Yes]
Remediation strategy:	



```
Activities  Terminal  Thu 06:29  ustudent@ubu-ustudent: ~  
File Edit View Search Terminal Help  
ustudent@ubu-ustudent:~$ dpkg -s rsyslog  
Package: rsyslog  
Status: install ok installed  
Priority: important  
Section: admin  
Installed-Size: 1432  
Maintainer: Ubuntu Developers <ubuntu-devel-discuss@lists.ubuntu.com>  
Architecture: amd64  
Version: 8.32.0-1ubuntu4  
Provides: linux-kernel-log-daemon, system-log-daemon  
Depends: libc6 (>= 2.27), libestr0 (>= 0.1.4), libfastjson4 (>= 0.99.7), libsystemd0 (>= 209), libuuid1 (>= 2.16), zlib1g (>= 1:1.1.4), init-system-helpers (>= 1.47~), lsb-base (>= 3.2-14), adduser, ucf  
Recommends: logrotate  
Suggests: rsyslog-mysql | rsyslog-pgsql, rsyslog-doc, rsyslog-gnutls, rsyslog-gssapi, rsyslog-relp, apparmor (>= 2.8.96~2541-0ubuntu4~)  
Conflicts: linux-kernel-log-daemon, system-log-daemon  
Conffiles:  
/etc/apparmor.d/usr.sbin.rsyslogd 1d7263e852f18f89b136525ba4033166  
/etc/default/rsyslog c7af23c348917e877399baea1849727a  
/etc/init.d/rsyslog bd41a0654a192d74dfb9c551b06fa855  
/etc/logcheck/ignore.d.server/rsyslog d77ab13caf836413b9debd906948d761  
/etc/logrotate.d/rsyslog 8159203bf98416c898849057f585e31f  
/etc/rsyslog.conf d1d930c884e1f403e821ec09e18cfd2e  
Description: reliable system and kernel logging daemon
```



Compliance Evaluation on Ubuntu

Benchmark:	4.2.1.3 Ensure logging is configured
Compliance:	[Yes]
Remediation strategy:	

```
Activities Terminal Thu 06:40 ●
ustudent@ubu-ustudent: ~
File Edit View Search Terminal Help
/etc/rsyslog.conf:13:module(load="imuxsock") # provides support for local system logging
/etc/rsyslog.conf:25:module(load="imklog" permitnonkernelfacility="on")
/etc/rsyslog.conf:35:$ActionFileDefaultTemplate RSYSLOG_TraditionalFileFormat
/etc/rsyslog.conf:38:$RepeatedMsgReduction on
/etc/rsyslog.conf:43:$FileOwner syslog
/etc/rsyslog.conf:44:$FileGroup adm
/etc/rsyslog.conf:45:$FileCreateMode 0640
/etc/rsyslog.conf:46:$DirCreateMode 0755
/etc/rsyslog.conf:47:$Umask 0022
/etc/rsyslog.conf:48:$PrivDropToUser syslog
/etc/rsyslog.conf:49:$PrivDropToGroup syslog
/etc/rsyslog.conf:54:$WorkDirectory /var/spool/rsyslog
/etc/rsyslog.conf:59:$IncludeConfig /etc/rsyslog.d/*.conf
/etc/rsyslog.d/20-ufw.conf:2:msg,contains,"[UFW " /var/log/ufw.log
/etc/rsyslog.d/50-default.conf:8:auth,authpriv.* /var/log/auth.log
/etc/rsyslog.d/50-default.conf:9:*. *;auth,authpriv.none -/var/log/syslog
/etc/rsyslog.d/50-default.conf:12:kern.* -/var/log/kern.log
/etc/rsyslog.d/50-default.conf:14:mail.* -/var/log/mail.log
/etc/rsyslog.d/50-default.conf:23:mail.err /var/log/mail.err
/etc/rsyslog.d/50-default.conf:39:*.emerg :omusrmsg:*
ustudent@ubu-ustudent:~$
```

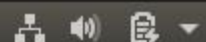



Compliance Evaluation on Ubuntu

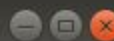
Benchmark:	5.2.1 Ensure permissions on <code>/etc/ssh/sshd_config</code> are configured
Compliance:	[No]
Remediation strategy:	Set ownership of <code>/etc/ssh/sshd_config</code> to <code>root:root</code> and restrict permissions to <code>600</code> .

Activities Terminal ▾

Thu 06:43 ●



ustudent@ubu-ustudent: ~



File Edit View Search Terminal Help

```
ustudent@ubu-ustudent:~$ stat /etc/ssh/sshd_config
```

```
File: /etc/ssh/sshd_config
```

```
Size: 3264          Blocks: 8          IO Block: 4096   regular file
```

```
Device: 801h/2049d Inode: 656964       Links: 1
```

```
Access: (0644/-rw-r--r--)  Uid: (  0/   root)   Gid: (  0/   root)
```

```
Access: 2026-06-25 04:07:09.420000000 -0400
```

```
Modify: 2018-02-09 21:31:46.000000000 -0500
```

```
Change: 2020-09-26 14:03:10.213275209 -0400
```

```
Birth: -
```

```
ustudent@ubu-ustudent:~$
```





Compliance Evaluation on Ubuntu

Benchmark:	<i>5.2.13 Ensure only strong Ciphers are used</i>
Compliance:	<i>[Yes]</i>
Remediation strategy:	

```
Activities  Terminal ▾  Thu 06:51 ●  [Network, Sound, Print, Power icons]
ustudent@ubu-ustudent: ~
File Edit View Search Terminal Help
ustudent@ubu-ustudent:~$ sudo sshd -T | grep -i '^ciphers'
[sudo] password for ustudent:
ciphers chacha20-poly1305@openssh.com,aes128-ctr,aes192-ctr,aes256-ctr,aes128-g
cm@openssh.com,aes256-gcm@openssh.com
ustudent@ubu-ustudent:~$
```




Compliance Evaluation on Ubuntu

Benchmark:	5.3.1 Ensure password creation requirements are configured
Compliance:	[No]
Remediation strategy:	Install libpam-pwquality , configure PAM to use pam_pwquality.so , and define password length and complexity settings in /etc/security/pwquality.conf .

```
Activities  Terminal  Thu 07:08  ustudent@ubu-ustudent: ~  
File Edit View Search Terminal Help  
ustudent@ubu-ustudent:~$ grep -E '\s*password\s+requisite\s+pam_pwquality\.so.*  
\bretry=3\b' /etc/pam.d/common-password  
ustudent@ubu-ustudent:~$ grep -E '^\s*(minlen|dcredit|ucredit|ocredit|lcredit|m  
inclass)\s*=' /etc/security/pwquality.conf  
ustudent@ubu-ustudent:~$
```



Section Three:

Access and Forensics

Access and Forensics

In this section, you'll continue your mission to secure NuttyUtility's digital landscape by diving into the critical areas of access permissions and forensic analysis. You'll evaluate who has access to specific directories on both Linux and Windows systems, ensuring proper control measures are in place. Additionally, you'll analyze packet capture (PCAP) files to uncover malicious network activities.

Your Task:

- Evaluate the permissions for specific folders on both Linux and Windows systems.
- Determine who has access and control over these folders and provide necessary commands to modify permissions.
- Analyze the provided PCAP files on both the Ubuntu and Windows VMs.
- Identify the source IP, targeted port, and any compromised user accounts or passwords.



Access and Permissions - Ubuntu

Use the **/home/ustudent/Documents/data** folder to answer the following questions

What is the permission to access the folder?	<i>[drw- rw- r--]</i>
Who can read the contents of the folder?	<i>[The owner and group members can read/write , and all other users can read/list the folder contents.]</i>
Who is the owner of the folder?	<i>[ustudent]</i>
What permission should I set so only the owner can read the folder?	<i>[drwx-----]</i>
Who can read the file inside the folder?	<i>[The owner, group members, and all other users can read the file, assuming they also have execute/traverse permission on the containing folder.]</i>



Access and Permissions - Windows

Use the C:\data folder to answer the following questions

Who has permission to access the folder?	<i>[student, Everyone, Administrators, System]</i>
Who has full control of the folder and its contents?	<i>[student, Everyone, Administrators, System]</i>
What command would you use to remove "Everyone" -s permissions?	<i>[icacls "C:\data" /remove "Everyone"]</i>
What command would you use to add read permissions to "user2" for the folder?	<i>[icacls "C:\data" /grant "user2:R"]</i>



Packet Analysis

Use the bruteforce2.pcap file on the Ubuntu VM to answer the following questions:

What was the source IP of the attack?	<i>[10.0.2.7]</i>
What port was targeted?	<i>[23]</i>
What password was used?	<i>[eSq_]</i>
Which user was compromised?	<i>[ubu-ustudent]</i>



Packet Analysis

Use the lateralmovement.pcap file on the Windows VM to answer the following questions:

What was the source IP of the attack?	<i>[10.0.2.7]</i>
Did the attacker try to access your machine from the same network?	<i>[Yes]</i>
What port was targeted?	<i>[445]</i>
What file did the attacker try to access?	<i>[payroll_20200927000951_1871.xls]</i>
Was the attacker able to access the file?	<i>[Yes]</i>



Section Four: Vulnerability Assessment

Vulnerability Assessment

In this segment, you'll transform into a cyber detective, using Nmap to perform detailed network scans on both Windows and Ubuntu VMs. Your mission is to identify open ports and vulnerabilities, providing critical insights into potential security risks. By leveraging Nmap NSE scripts, you'll uncover and address vulnerabilities, ensuring our newly acquired systems are secure and ready for integration with NuttyUtility's network.

Your Task:

You need to perform the following tasks on the provided VMs:

- Use Nmap to identify open ports and provide screenshot evidence.
- Use an Nmap NSE script, like `nmap-vulners`, to scan for vulnerabilities and provide screenshot evidence of any critical vulnerabilities found.
- Choose one critical vulnerability from the scan results and provide its CVE, CVSS score, a brief description, and a remediation strategy.



Open ports

Use Nmap to identify the open ports on the Windows VM, and provide a screenshot of the results.

```
ustudent@ubu-ustudent: ~/Desktop/pcaps
File Edit View Search Terminal Help
ustudent@ubu-ustudent:~/Desktop/pcaps$ sudo nmap -sS -sV -Pn -T4 10.0.2.4
[sudo] password for ustudent:

Starting Nmap 7.60 ( https://nmap.org ) at 2026-06-25 11:20 EDT
Nmap scan report for 10.0.2.4
Host is up (0.0020s latency).
Not shown: 990 closed ports
PORT      STATE SERVICE          VERSION
7/tcp     open  echo
9/tcp     open  discard?
13/tcp    open  daytime         Microsoft Windows USA daytime
17/tcp    open  qotd             Windows qotd (English)
19/tcp    open  chargen
80/tcp    open  http             Microsoft IIS httpd 10.0
135/tcp   open  msrpc            Microsoft Windows RPC
139/tcp   open  netbios-ssn      Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds     Microsoft Windows 7 - 10 microsoft-ds (workgroup: WORKGROUP)
3389/tcp  open  ms-wbt-server    Microsoft Terminal Services
MAC Address: 08:00:27:68:CF:18 (Oracle VirtualBox virtual NIC)
Service Info: Host: WIN10-USTUDENT; OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 159.23 seconds
ustudent@ubu-ustudent:~/Desktop/pcaps$
```



Open ports

Use Nmap to identify the open ports on the Ubuntu VM, and provide a screenshot of the results.

The screenshot shows a terminal window titled "ustudent@ubu-ustudent: ~/Desktop/pcaps" with a menu bar (File, Edit, View, Search, Terminal, Help). The terminal displays the command `sudo nmap -sS -sV -Pn -T4 10.0.2.3` and its output. The output indicates the host is up and lists 11 open ports with their corresponding services and versions. A service fingerprint (SF) is also provided for port 17/TCP.

```
ustudent@ubu-ustudent: ~/Desktop/pcaps
ustudent@ubu-ustudent:~/Desktop/pcaps$ sudo nmap -sS -sV -Pn -T4 10.0.2.3

Starting Nmap 7.60 ( https://nmap.org ) at 2026-06-25 11:33 EDT
Nmap scan report for ubu-ustudent (10.0.2.3)
Host is up (0.0000060s latency).
Not shown: 991 closed ports
PORT      STATE SERVICE      VERSION
13/tcp    open  daytime
17/tcp    open  qotd?
21/tcp    open  ftp          vsftpd 2.0.8 or later
22/tcp    open  ssh          OpenSSH 7.6p1 Ubuntu 4 (Ubuntu Linux; protocol 2.0)
23/tcp    open  telnet       Linux telnetd
37/tcp    open  time         (32 bits)
80/tcp    open  http         Apache httpd 2.4.29 ((Ubuntu))
139/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
1 service unrecognized despite returning data. If you know the service/version,
please submit the following fingerprint at https://nmap.org/cgi-bin/submit.cgi
?new-service :
SF-Port17-TCP:V=7.60%I=7%D=6/25%Time=6A3D4A3E%P=x86_64-pc-linux-gnu%r(NULL
SF:;,34,"If\x20you\x20sow\x20your\x20wild\x20oats,\x20hope\x20for\x20a\x20c
SF:rop\x20failure\.\n");
Service Info: Host: Welcome; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nma
p.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 13.42 seconds
ustudent@ubu-ustudent:~/Desktop/pcaps$
```




NSE Script scan

Use an Nmap NSE script - [nmap-vulners](#) would be good for this - to scan for vulnerabilities on the Ubuntu VM and provide a screenshot of a screenshot front he results, that shows a critical vulnerability.

```
Activities Terminal Thu 12:14 ●
ustudent@ubu-ustudent: ~
File Edit View Search Terminal Help
new-service :
SF-Port17-TCP:V=7.60%I=7%D=6/25%Time=6A3D5226%P=x86_64-pc-linux-gnu%r(NULL
SF: ,26,"You\x20will\x20be\x20run\x20over\x20by\x20a\x20beer\x20truck\.\n");
Service Info: Host: Welcome; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Host script results:
| smb-vuln-cve-2017-7494:
|   VULNERABLE:
|     SAMBA Remote Code Execution from Writable Share
|       State: LIKELY VULNERABLE
|       IDs:   CVE:CVE-2017-7494
|       Risk factor: HIGH  CVSSv3: 7.5 (HIGH) (CVSS:3.0/AV:N/AC:H/PR:L/UI:N/S:U/C
|:H/I:H/A:H)
|       All versions of Samba from 3.5.0 onwards are vulnerable to a remote
|       code execution vulnerability, allowing a malicious client to upload a
|       shared library to a writable share, and then cause the server to load
|       and execute it.
|
|       Disclosure date: 2017-05-24
|       Check results:
|         Samba Version: 3.X - 4.X
|         Writable share found.
|         Name: \\10.0.2.3\data
|         File written to remote share, but unable to execute payload either due
| to unknown actual path, or the system may be patched.
|       Extra information:
|         All writable shares:
|           Name: \\10.0.2.3\data
|       References:
```



Vulnerability Analysis

Choose one **critical vulnerability** from the scan results, and do the following tasks.

What is the CVE and CVSS score of the vulnerability you choose?

[**9.8 Critical** under CVSS v3.1 according to NVD.]

Describe this vulnerability in 1-3 sentences.

[CVE-2017-7494 is a **Samba remote code execution vulnerability** affecting Samba versions from **3.5.0** up to vulnerable 4.4.x, 4.5.x, and 4.6.x releases. A malicious client can upload a shared library to a writable Samba share and cause the **smbd** server to load and execute it.]

Provide a remediation strategy for this vulnerability.

[Upgrade Samba to a fixed version: **4.6.4, 4.5.10, 4.4.14, or later**, or apply the vendor security patch for your Linux distribution. As a temporary workaround, Samba advised adding **nt pipe support = no** to the **[global]** section of **smb.conf** and restarting **smbd**, but this may break some Windows client functionality]